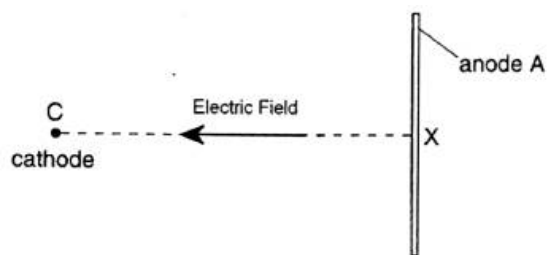


2 (a) The *electric field strength* at a point is the electric force acting per unit positive charge placed at that point.

Note: Common mistake such as not referring to a positive charge or not including the term 'per' in the definition.

(b) (i)



(ii) The electric field strength at the mid-point between C and X is $E = 2.4 \text{ kV m}^{-1}$ or 2400 N C^{-1}

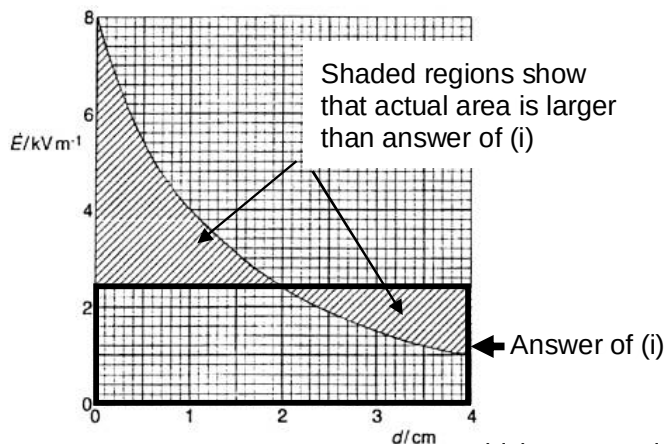
$$\begin{aligned} F &= qE \\ &= eE \\ &= (1.60 \times 10^{-19})(2400) \\ &= 3.84 \times 10^{-16} \text{ N} \end{aligned}$$

(c) (i) Assuming a constant force F , the work done by the field to accelerated the electron from C to X is $W = Fd$

$$\begin{aligned} &= (3.84 \times 10^{-16})(0.040) \\ &= 1.536 \times 10^{-17} \text{ J} \end{aligned}$$

$$\begin{aligned} \text{The potential difference } V &= \frac{W}{q} \\ &= \frac{(1.536 \times 10^{-17})}{(1.60 \times 10^{-19})} \\ &= 96 \text{ V.} \end{aligned}$$

(ii) The magnitude of the potential difference calculated in (i) is an under-estimate. The average electric field shown in Fig. 2.2 is greater than that used in part (i). The actual potential difference is the area under the field-distance graph.



3 (a) A progressive wave is a travelling disturbance, in which energy is propagated through a distance by a series of successive vibrations